MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 6130

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen

See figure 1

Mounting position: Any

Weight: 3-ounces nominal (85 grams)

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Ebb	Ec	egx	egy	ib
Unit:	Vac	kv	kv	Vdc	Vdc	v	v	a
Maximum:	6.6	3.0	3.0	---	---	200	---	35
Minimum:	5.7	---	---	800	---	---	---	---
		(Note 1)	(Note 2)				(Note 3)	
TEST CONDITIONS:	6.3	3.0	---	---	0	---	130	---

ABSOLUTE RATINGS:

Parameter:	Ib	tk	$\frac{dik}{dt}$	Pb	TA	prr	Barometric pressure, reduced
Unit:	mAdc	sec	a/μs	---	°C	pps	mmHg
Maximum:	45	---	750	0.3×10^9	90	---	87.0
Minimum:	---	120	---	---	-50	---	---
TEST CONDITIONS:	---	120	---	---	---	2,800	---

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3241	Heater current	-		0.65	II	If	2.0	2.5	Aac
3267	Instantaneous starting	5,7	epy = 3,000 v (min)	0.65	II	---	---	---	---
(D) 3247	DC anode voltage for conduction	8		0.65	II	Ebb	---	300	Vdc
3246	Operation (1)	5,9,10	epy = 4,000 v (min)	0.65	II	egy	---	130	v
(D) 3251	Pulse emission (method A)	-	ik = 35 a (min); prr = 60 ±10%; tp = 5.0 µs ±10%; tr = 0.5 µs (max); specified time interval = 2.5 µs	0.65	II	egk	---	150	v
	Quality conformance inspection, part 2								
1036	Bump	-	Angle = 20°	---	---	---	---	---	---
3256	Anode delay time	5,9	Operation (1); t = 120 seconds	---	---	tad	---	0.6	µs
3256	Anode delay time drift	11	Anode delay time	---	---	Δtad	---	0.15	µs
3261	Time jitter	5	epy = 1,500 v (max)	---	---	tj	---	0.02	µs
3246	Operation (2)	5,6	t = 5.0 minutes	---	---	egy	---	130	v
	Quality conformance inspection, part 3								
---	Life test	5	Group B; t = 500 hours; t = 96 hours "on" and 1 hour "off" (tube mounted horizontally)	---	---	---	---	---	---
---	Life-test end points:								
3246	Operation (1)	-		---	---	egy	---	140	v
3247	DC anode voltage for conduction	-		---	---	Ebb	---	750	Vdc
3261	Time jitter	-		---	---	tj	---	0.04	µs
(D) 1031	Low-frequency vibration	4,12	No voltages; F = 12 to 50 Hz	---	---	---	---	---	---
(D) 1031	High-frequency vibration	4,12	t = 30 seconds (min)	---	---	---	---	---	---
(D) 3246	Operation at elevated temperature	5,12	t = 5.0 hours; TA = 90°C	---	---	egy	---	130	v
3246	Operation (3)	5,6,12	t = 5.0 hours	---	---	egy	---	130	v

NOTES:

1. For instantaneous starting applications where anode voltage is applied instantaneously, the power supply filter design shall be such that the maximum permissible epy is 3,000 v and shall not be attained in less than 0.04 second.

2. In pulsed operation, the peak inverse voltage, exclusive of a spike of 0.05 µs maximum duration, shall not exceed 1,500 volts during the first 25 µs after the pulse.

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NOTES: -Continued

3. Driver pulse, measured at tube socket with thyratron grid disconnected; egy = 175 v (min), time of rise = 0.5 μs (max), grid pulse duration = 2.0 μs (min). Impedance of drive circuits = 1,500 ohms (max).

4. There shall be no pronounced resonance in the specified range.

5. The circuit constants shall be chosen so that at epy = 3.0 kv under resonant charging conditions, dik/dt = 750 a μs (min), ib = 35 a, tp = 0.5 μs ±10 percent, prr = 3,000. The grid pulse characteristics shall be tp = 2.0 μs (max), tr = 0.5 μs (min), driver impedance = 1,500 ohms (min).

6. The tube shall operate satisfactorily in an evacuated chamber in which the pressure does not exceed 70 mmHg.

7. The tube shall operate satisfactorily on push button starting within three attempts when the anode voltage epy is applied to the tube under test in such a manner as to rise from 0 to 3,000 v within 0.03 second. The filter in the rectifier shall be designed so that epy reaches at least 1,500 v within 0.015 second.

8. This test shall be conducted within 60 seconds after the operation (1) test.

9. The tube shall operate continuously for 5 minutes without evidence of arc-back or anode heating.

10. This test is to be the first test performed at the conclusion of the holding period.

11. During the interval between 2 minutes and 7 minutes of the anode delay time test, the change in anode delay time (Δtad) relative to the tad value observed on the anode delay time test shall not exceed the specified value.

(D) 12. This test shall be performed annually. A sample size of three tubes shall be used, with no failures allowed. In case of a sample failure, the test shall become a quality conformance inspection (QCI), part 2 test, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3 test.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

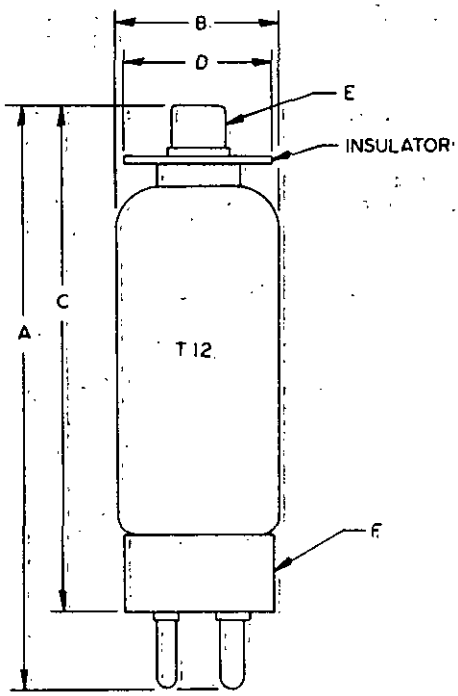
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2878)

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PIN CONNECTIONS	
Pin No.	Element
1	h
2	k
3	g
4	h, k
cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	---	5.188 (131.78)
B	---	1.563 (39.70)
C	4.188 (106.38)	4.563 (115.90)
Quality conformance inspection, part 3 (see note)		
D	---	1.469 (37.31)
E	Cap: C1-1 (EIA)	
F	Base: A4-9 (EIA)	

Ⓓ

NOTE:
A sample size of three tubes shall be checked annually, with no failures allowed. In case of a sample failure, the test shall become a quality conformance inspection (QCI), part 2 test, for three successful consecutive submissions at which time the test may revert to the QCI, part 3 test.

FIGURE 1. Outline drawing of electron tube type 6130.

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